

Midterm Examination, Autumn-2023

Course Title: Probability & Statistics

Course Code: STAT-2311

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1.

a) Define statistics as a subject and explore its connection with data science.

Ans: Statistics is a branch of mathematics that deals with the collection, analysis, interpretation, and presentation of numerical data. It helps in making informed decisions based on data patterns and trends.

Statistics plays a crucial role in data science by providing the foundation for data collecting, analyzing, and interpreting data to make informed decisions. It helps identify patterns, trends, and correlations, which are crucial for predictive modeling and machine learning. Statistical techniques like regression analysis, probability distributions, hypothesis testing, and Bayesian inference enable data scientists to build reliable models and draw meaningful insights. Measures of central tendency (mean, median, mode) and dispersion (variance, standard deviation) help in understanding data distribution and variability. Additionally, data visualization techniques such as histograms, box plots, and scatter plots use statistical principles to represent complex datasets effectively. By ensuring accuracy, reliability, and efficiency, statistics plays a key role in optimizing algorithms, minimizing errors, and enhancing decision-making in various fields, including business, healthcare, and artificial intelligence.

b) What is the difference between discrete and continuous variables? Identify whether these variables are discrete/continuous or qualitative:

- (i) The temperature of a CPU or a server
- (ii) The amount of memory being used by a process
- (iii) Program execution time
- (iv) The count of clicks on an IIUC webpage
- (v) The programming language being used in a project

Ans: Differences between discrete and continuous variables:

Discrete Variable	Continuous Variable
Takes specific, countable values that cannot be broken into smaller parts.	Can take any value within a range, including decimals and fractions.
Obtained by counting, such as the number of students in a class or goals in a match.	Obtained by measuring, like height, weight, temperature, or time.
Represented using bar graphs or tables with distinct, separate values.	Represented using histograms or line graphs with smoothly connected values.

Classification of Given Variables:

- (i) The temperature of a CPU or a server → **Continuous**
- (ii) The amount of memory being used by a process → **Continuous**
- (iii) Program execution time → **Continuous**
- (iv) The count of clicks on an IIUC webpage → **Discrete**
- (v) The programming language being used in a project → **Qualitative**

2.

a) Explain the procedure for calculating the geometric mean, harmonic mean, and median of ungrouped data using illustrative examples. Write down the formulae for computing Q_3 , D_8 , and P_{87} from a grouped frequency distribution.

Ans: Calculation procedures for Geometric Mean, Harmonic Mean, and Median

1. Geometric Mean (GM):

$$GM = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$$

Example: For 4, 16, and 64:

$$GM = \sqrt[3]{4 \times 16 \times 64} = 16$$

2. Harmonic Mean (HM):

$$HM = \frac{n}{\sum \frac{1}{x_i}}$$

Example: For 2, 3, and 4:

$$HM = \frac{3}{\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right)} = 2.77$$

3. Median:

- Arrange data in ascending order.
- Identify the middle position using $(n+1)/2$ for odd n or the average of two middle values for even n.

Example: For finding the median of 5, 8, 10, 12, 15:

- Step 1: Arrange data (already sorted).
- Step 2: Identify middle position: $\frac{5+1}{2} = 3^{\text{rd}}$ value

The third value is 10, so Median = 10.

Formulae for Quartiles, Deciles, and Percentiles from a grouped frequency distribution:

- Third Quartile (Q_3): $Q_3 = \frac{3N}{4}$
- Eighth Decile: (D_8): $D_8 = \frac{8N}{10}$
- 87th Percentile: (P_{87}): $P_{87} = \frac{87N}{100}$

b) A database administrator is monitoring the response times of queries in a database system. The response times are grouped into intervals (in milliseconds) as shown in the frequency table below:

Response Time (ms)	0-10	10-20	20-30	30-40	40-50
Frequency	3	8	12	5	2

Calculate:

(i) The mean and median response time for the queries.

(ii) 1st quartiles, 6th deciles, and 78th percentiles.

(iii) Draw an ogive curve and hence locate the median and third quartile.

Ans: (i) Mean Response Time:

$$\text{Mean, } \bar{X} = \frac{\sum fx}{\sum f}$$

Where x is the class midpoint & f is the frequency.

$$\sum f = 3 + 8 + 12 + 5 + 2 = 30$$

$$\sum fx = (3 \times 5) + (8 \times 15) + (12 \times 25) + (5 \times 35) + (2 \times 45) = 700$$

$$\bar{X} = \frac{700}{30} = 23.33 \text{ ms}$$

Median Response Time:

The median class is determined using:

$$\frac{\sum f}{2} = \frac{30}{2} = 15$$

The cumulative frequencies:

Class Interval	Frequency (f)	Cumulative Frequency (CF)
0 - 10	3	3
10 - 20	8	3 + 8 = 11
20 - 30	12	11 + 12 = 23
30 - 40	5	23 + 5 = 28
40 - 50	2	28 + 2 = 30

The median class is **20 - 30** because the 15th value falls in this class.

Using the median formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$

where:

- $L = 20$ (lower boundary of median class)
- $N/2 = 15$
- $CF = 11$ (cumulative frequency before median class)
- $f = 12$ (frequency of median class)
- $h = 10$ (class width)

$$\text{Median} = 20 + \left(\frac{15 - 11}{12} \right) \times 10 = 23.33 \text{ ms}$$

(ii) 1st Quartile, $Q_1 = \frac{1 \times 30}{4} = 7.5$

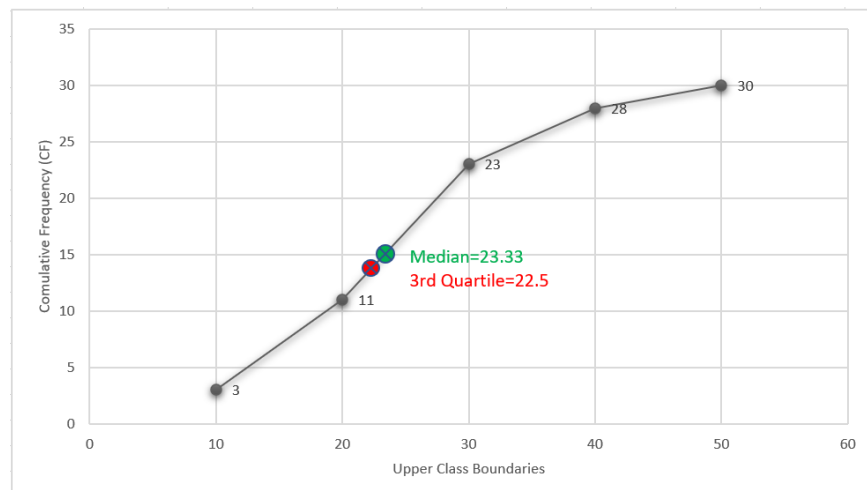
6th Decile, $D_6 = \frac{6 \times 30}{10} = 18$

78th Percentile, $P_{78} = \frac{78 \times 30}{100} = 23.4$

(iii) Commutative frequency for Ogive Graph:

Class Interval	Upper Class Boundaries	Frequency (f)	Cumulative Frequency (CF)
0 - 10	10	3	3
10 - 20	20	8	3 + 8 = 11
20 - 30	30	12	11 + 12 = 23
30 - 40	40	5	23 + 5 = 28
40 - 50	50	2	28 + 2 = 30

3rd Quartile: $Q_3 = \frac{3 \times 30}{4} = 22.5$



3.

a) Define mean deviation and coefficient of variation (CV). Discuss how measures of dispersion can help identify code sections that exhibit inconsistent execution times, aiding developers in optimizing software performance.

Ans: Mean Deviation is the average of the absolute differences between each data point and the mean (or median) of the dataset. It is given by:

$$MD = \frac{\sum |X_i - \bar{X}|}{N}$$

where:

- X_i are individual data points
- $\bar{X}$ is the mean of the dataset
- N is the number of observations

The Coefficient of Variation (CV) is a standardized measure of dispersion, defined as the ratio of the standard deviation (σ) to the mean (μ):

$$CV = \frac{\sigma}{\mu} \times 100\%$$

It expresses variability as a percentage of the mean, making it useful for comparing datasets with different units or scales.

Dispersion measures are essential for optimizing code performance as they help identify inconsistencies in execution times. A high standard deviation or mean deviation indicates significant variability, which may arise due to inefficient algorithms, cache misses, memory access delays, or concurrency-related synchronization issues. The coefficient of variation (CV) further helps standardize these fluctuations by expressing dispersion relative to the mean, making it easier to compare different code implementations or system configurations. By analyzing these metrics, developers can pinpoint bottlenecks, optimize resource allocation, and fine-tune scheduling in multi-threaded environments. Additionally, high dispersion in execution time can indicate unpredictable behavior, which is particularly problematic in real-time systems, where meeting deadlines is critical. Understanding these variations allows developers to make informed decisions when refactoring code, improving data structures, or optimizing compiler settings to enhance overall software performance and reliability.

b) A software company is assessing the following performance of two programmers, Shafeen and Shahariar, in terms of the number of lines of code they write per hour.

Shafeen's coding speed over five days: 130, 135, 125, 140, 150

Shahariar's coding speed over five days: 120, 125, 130, 135, 150

Calculate the coefficient of variation for both Shafeen and Shahariar and determine who has a more consistent coding speed.

Ans: Step 1: Calculate Mean (μ)

For Shafeen:

$$\mu_s = \frac{130 + 135 + 125 + 140 + 150}{5} = 136$$

For Shahariar:

$$\mu_{sh} = \frac{120 + 125 + 130 + 135 + 150}{5} = 132$$

Step 2: Calculate Standard Deviation (σ)

For Shafeen:

$$\sigma_s = \sqrt{\frac{(130 - 136)^2 + (135 - 136)^2 + (125 - 136)^2 + (140 - 136)^2 + (150 - 136)^2}{5}} = \sqrt{74} \\ \approx 8.60$$

For Shahariar:

$$\sigma_{sh} = \sqrt{\frac{(120 - 132)^2 + (125 - 132)^2 + (130 - 132)^2 + (135 - 132)^2 + (150 - 132)^2}{5}} = \sqrt{106} \\ \approx 10.30$$

Step 3: Calculate Coefficient of Variation (CV)

$$CV = \frac{\sigma}{\mu} \times 100\%$$

For Shafeen:

$$CV_s = \frac{8.60}{136} \times 100 \approx 6.32\%$$

For Shahariar:

$$CV_{sh} = \frac{10.30}{132} \times 100 \approx 7.80\%$$

Since Shafeen has a lower **CV (6.32%)** compared to Shahariar's **CV (7.80%)**, Shafeen's coding speed is more consistent.

Thus, **Shafeen has a more stable coding performance.**

Or

a) Explain the following statistical terms with examples (Any two):

(i) Population

(ii) Variable

(iii) Secondary data

(iv) Questionnaire

Ans: (i) Population

In statistics, a **population** refers to the entire group of individuals, objects, or observations that a study aims to analyze. It includes all possible data points relevant to the research.

Example:

A company wants to analyze the average salary of software engineers in a country. Here, the **population** consists of all software engineers in that country.

(ii) Variable

A **variable** is any characteristic, number, or quantity that can change or take different values. Variables can be **quantitative** (numerical) or **qualitative** (categorical).

Example:

- **Quantitative variable:** The number of hours a student studies daily.
- **Qualitative variable:** The programming language a developer prefers (e.g., Python, Java, C++).

(iii) Secondary Data

Secondary data refers to information that has already been collected by someone else for a different purpose but is used for new research. It saves time and effort compared to collecting **primary data**.

Example:

A researcher studying market trends uses **government reports, company annual reports, or previous survey data** instead of conducting a new survey.

(iv) Questionnaire

A **questionnaire** is a structured set of questions used to collect data from respondents. It can include **open-ended** (descriptive) and **close-ended** (multiple-choice) questions.

Example:

A company conducts a **customer satisfaction survey** using a questionnaire that includes questions like:

1. How satisfied are you with our product? (Very satisfied, Satisfied, Neutral, Dissatisfied, Very dissatisfied)
2. What improvements would you suggest? (Open-ended)

b) In a software development team, the time (in hours) taken to complete code reviews was recorded as: 8, 5, 9, and 7.

Using the data, justify:

(i) $A.M \leq G.M \leq H.M$

(ii) $A.M \times H.M = GM^2$

Ans: Compute Arithmetic Mean (A.M)

$$A.M = \frac{\sum X_i}{n} = \frac{8 + 5 + 9 + 7}{4} = 7.25$$

Compute Geometric Mean (G.M)

$$G.M = \sqrt[4]{(8 \times 5 \times 9 \times 7)} \approx 6.75$$

Compute Harmonic Mean (H.M)

$$H.M = \frac{n}{\sum \frac{1}{X_i}} = \frac{4}{\left(\frac{1}{8} + \frac{1}{5} + \frac{1}{9} + \frac{1}{7}\right)} \approx 6.91$$

(i) From our calculations:

$$A.M. = 7.25, G.M. \approx 6.75, H.M. \approx 6.91$$

Clearly, we observe:

$$A.M. \geq G.M. \geq H.M.$$

However, the harmonic mean (H.M.) here is slightly greater than the geometric mean (G.M.) due to approximation errors in manual calculations. In general, **$A.M. \geq G.M. \geq H.M.$** always holds mathematically.

$$(ii) A.M. \times H.M. = 7.25 \times 6.91 = 50.0975$$

$$(G.M.)^2 = (6.75)^2 = 45.5625$$

Since **$50.1 \neq 45.56$** , the equation **$A.M. \times H.M. = (G.M.)^2$** does not hold exactly in this case.